

Oil Shock Racing – Project Report

1. Introduction

The **Oil Shock Racing** project is a simulation-based platform designed to model oil price shocks and analyze their impact on financial markets. Oil prices play a crucial role in influencing inflation, sector performance, and macroeconomic stability. This project provides an interactive environment where users can understand these dynamics through simulation and decision-making.

2. Objective

The main objectives of this project are:

- To simulate oil market shocks (supply, demand, and policy shocks)
 - To analyze the impact of these shocks on oil prices, inflation, and sectors
 - To visualize market behavior using interactive charts
 - To develop a gamified system where users make portfolio decisions
 - To evaluate decision-making through a scoring system
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3. System Architecture

The system follows a full-stack architecture:

- **Frontend:** React.js (User Interface and Visualization)
- **Backend:** Node.js with Express (API layer)
- **Simulation Engine:** Python (Shock generation and market modeling)

Workflow:

- User triggers simulation from frontend
 - Frontend calls backend API
 - Backend executes Python simulation
 - Python returns market data
 - Backend sends response to frontend
 - Frontend visualizes data and updates UI
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4. Technologies Used

- **Frontend:** React.js, Axios, Recharts
 - **Backend:** Node.js, Express.js
 - **Simulation:** Python
 - **Visualization:** Recharts library
 - **Tools:** VS Code, PowerShell
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5. Methodology

5.1 Shock Generation

The system simulates three types of shocks:

Supply Shock: Caused by disruptions like wars or production cuts, leading to price increase

Demand Shock: Caused by economic slowdown, leading to price decrease

Policy Shock: Caused by regulatory or OPEC decisions

5.2 Market Simulation

The simulation calculates:

- Oil Price = Base Price + Shock Impact
- Inflation = 3% of price change
- Energy Sector = Positive correlation with oil price
- Airline Sector = Negative correlation with oil price

5.3 Visualization

- Time-series price data is plotted using line charts
- Each simulation updates the dataset dynamically
- Trends are displayed in real-time

5.4 Game Logic

- User starts with a fixed portfolio (cash + stocks)
- User can buy energy or airline stocks

Based on shock type, decisions affect score:

- Supply shock → Energy performs better
 - Demand shock → Airline performs better
 - Score reflects decision quality
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6. Features

- Oil shock simulation engine
 - Real-time data visualization
 - Interactive dashboard
 - Portfolio management system
 - Decision-based scoring system
 - Leaderboard display
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7. API Design

Endpoint:

GET /api/simulate

Response:

- Shock type
 - Impact value
 - Updated oil price
 - Inflation
 - Sector impacts (energy, airline)
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8. Results

The system successfully:

- Simulates realistic oil shock scenarios
 - Demonstrates how shocks affect markets
 - Allows users to make strategic decisions
 - Visualizes trends through charts
 - Provides feedback via scoring system
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9. Limitations

- Uses simplified economic assumptions
 - No real-time market data integration
 - Linear relationships between variables
 - Basic game logic
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10. Future Enhancements

- Integration with real-time APIs (oil price data)
 - Advanced economic modeling
 - Machine learning-based prediction
 - Improved UI/UX design
 - Multiplayer leaderboard system
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11. Conclusion

The Oil Shock Racing project successfully demonstrates the impact of oil price shocks on financial markets using simulation and visualization. The gamified approach enhances user engagement and provides a practical understanding of economic concepts. This project showcases full-stack development skills along with data analysis and modeling capabilities.

12. References

- Energy market concepts and macroeconomic principles
 - Python documentation (NumPy, random)
 - React and Recharts documentation
 - Node.js and Express documentation
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